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General Comments

Candidates generally found the paper to be one in which they had the chance to show what they had learned.

Candidates should be reminded to follow the instructions given on the front of the question paper concerning the degree of accuracy required in answers. A significant proportion of candidates did not and this resulted in the loss of marks especially in **Questions 7(iii)**, **10(ii)** and **11.**

Candidates should also be reminded of the need for clear presentation in their work so that Examiners can see what they are trying to achieve. Any question requiring candidates to 'show that...' will not gain full marks if candidates appear to miss out necessary steps. In a similar way, method marks, available even if an answer is incorrect, will not be awarded if the method is not clear. The most common example of this occurs if a candidate makes an error forming a quadratic equation. If he/she then factorises and solves, or shows use of the quadratic formula the method mark(s) are earned, but if only the answers are given, no method mark is possible.

Comments on specific questions

Question 1

The majority of candidates knew to rewrite sec$x$ in terms of cos$x$ and to form a single fraction. They then used the identity $\cos^2 x + \sin^2 x = 1$ and most managed to reach the desired answer. A good knowledge of trigonometric relationships was displayed by these candidates whereas others used incorrect relationships and made little or no progress (a problem which also arose in **Question 11**). Some candidates worked from the right hand side successfully.

Question 2

(i) Many candidates knew exactly how to answer this part. Others were unsure of how to approach it. Some candidates used combinations rather than permutations or simple factorials (of 4, of 7 or of 8). Some favoured the approach of successive boxes i.e. 7 choices for the first box, 6 for the second etc. and thus calculating 7 x 6 x 5 x 4.

(ii) The majority of those who dealt correctly with part **(i)** managed this part.

The box method was more popular here: 4 choices (of even number) in the first box, 6 choices in the second etc. i.e. 4 x 6 x 5 x 4. Otherwise candidates worked out ${}^6P_3$ and multiplied this by the number of different even numbers.

Answers: **(i)** 840 **(ii)** 480

Question 3

Two valid methods of solution were seen (or even a combination of both). The more popular of these was to equate the two equations to obtain a new (quadratic) equation in x (and involving m). The use of $b^2 - 4ac$, because the line was a tangent and therefore there were equal roots, was common and the solution of $b^2 - 4ac = 0$ to give values of m was well handled by most.

The other method was to differentiate the quadratic and use the result $2x + 12$ in place of m in the line equation. This gave solutions for x and substitution was necessary to find m values, a fact which some candidates overlooked.

A number of candidates incorrectly offered ranges for their answer e.g. $4 < m < 20$. A common wrong method was to solve $x^2 + 12 + 18 = 0$ and then to find m.

Answer: $m = 4, 20$

Question 4

Although some candidates wrongly evaluated f(–2) and f(1) the vast majority knew to use f(2) and f(–1). The difficulty for some candidates arose in trying to link the two remainders too early. Those who obtained the two remainders independently of each other and then considered the '5 times' relationship usually went on to a correct answer. Some candidates formed equations in k and, for example, m (remainders of m and $5m$) and found the remainders (–9 and –45) before finding k by substitution.

As usual with this type of question f(2) = 0 and f(–1) = 0 were often seen. In many cases the '=0' was subsequently ignored, but if f(2) = 0 was solved the whole question became meaningless.

Answer: $k = -10$.

Question 5

Some candidates were quite comfortable with logarithms but many would benefit from more practice on this topic. The most common error was to see $\log_3(2a - b)$ written as $\log_3 2a - \log_3 b$.

The candidates who obtained the equations $a = b^2$ and $2a - b = 3$ were usually able to solve correctly but very few realised that $a = 1$, $b = -1$ was not a possible solution.

Answer: $a = \frac{9}{4}, b = \frac{3}{2}$

Question 6

Varying methods of solution were seen, not all fully supported by working. The most common involved the use of the factor theorem to obtain one solution and subsequent work to find a quadratic factor. This latter appeared either as long division, synthetic division or as comparison of coefficients – the linear factor being multiplied by $ax^2 + bx + c$. Candidates need to read this type of question carefully to check whether they are being asked for factors or a solution. Some candidates stopped short of the answer required merely providing three linear factors.

Answer: $x = -4, -\frac{1}{3}, 2$

Question 7

(i) Because $y = 3x - 5$ cuts the x-axis at a non-integer value those candidates who went straight to the construction of a table of values finished up with a graph which itself never touched the x-axis. Those candidates who realised what was going on lightly drew the line $y = 3x - 5$ and then reflected, in the x-axis, the part of the original line below the axis. Again, marks were often lost due to candidates not answering what had been asked for – in this case marking the coordinates where the graph met the axes.

(ii) This was well done.

(iii) With a required degree of accuracy of 3 significant figures the intersection of the two graphs would never be accurate enough, although some credit was given for this method.

Candidates who understood the modulus function considered the two equations $8x = 3x - 5$ (giving $x = -1$) and $8x = -(3x - 5)$ leading to the correct solution $\frac{5}{11}$. Of those candidates using algebra, many omitted the second possibility thereby ignoring the modulus function altogether. As with other questions (e.g. **Questions 5** and **11**) candidates often failed to give any thought to their answer offering both $\frac{5}{11}$ and -1 even though their graphs clearly showed that there was no intersection at -1.

Answers: **(iii)** $x = -\frac{5}{11}$ (or 0.455).

Question 8

(a)(i) Many candidates seemed uncertain of how to answer this question. The mathematical language and notation used here is an important part of the syllabus, and candidates should be prepared for this type of question. Of the efforts seen some included 'differential = 0' while others involved a table of values (only of use if the least value occurs at an integral value).

(ii) Very few candidates understood what was needed here and, of course, those with no answer in part **(i)** could not attempt this. Candidates should realise that a one-one relationship was necessary for the inverse to exist. This required a domain consisting of a continuous set of numbers on one side or the other of $x = -2$. Even those candidates who had some idea let themselves down by writing, for example, $f(x) \geq 0$ instead of $x \geq 0$.

(b)(i) This was correctly attempted by all but the weakest candidates although many, when dealing with the 2 in the denominator, forgot to multiply the 1 by this.

(ii) Some candidates could not deal correctly with $gh(x)$, some actually finding $hg(x)$ and others the product of $g(x)$ and $h(x)$. The majority, however, managed to find $gh(x)$ correctly and, by equating to $g^{-1}(x)$, formed a quadratic equation which they solved.

Answers: **(a) (i)** Least value = –10 when x = –2 **(ii)** Any suitable domain that makes f a one-one function (e.g. $x \geq -2$) **(b) (i)** $g^{-1}(x) = 2(x+1)$ **(ii)** $x = -1$ or 6.

Question 9

(a) Candidates were expected to answer this part by expanding the bracket and then integrating term by term. A large number of those trying to do this had an x^2 term and a constant term but no x term. There were also many candidates who had a constant term of –9. Candidates would benefit from more work on expanding similar types of expression.

(b)(i) Differentiation of a product was required here and this was handled well by the majority of candidates.

(ii) This clearly stated 'Hence...'. and so candidates needed to use their result from part **(i)** in this part. Candidates who offered an answer to part **(i)** in an unsimplified form now had to attempt to reduce it to one term and to compare with the integral here. Any candidate with a wrong answer in part **(i)** would have found this impossible and thus should have realised that their previous answer was incorrect.

Answers: **(a)** $\frac{3}{5}x^{\frac{5}{3}} - \frac{9}{2}x^{\frac{4}{3}} + 9x + c$ **(b)(i)** $\sqrt{x^2+6} + x\left(\frac{2x}{2\sqrt{x^2+6}}\right)$ **(ii)** $\frac{1}{2}x\sqrt{x^2+6} + c$

Question 10

This was a demanding question, involving use of logarithms, exponentials, differentiation of product or quotient, knowledge of how s, v and a are connected and how to deal with a time interval. A few candidates could cope with the whole question but some could not make any meaningful progress at all.

(i) It was expected that something similar to $e^5 = t^2 + 1$ with subsequent work would be seen. However, this was only seen occasionally and common errors included the misuse of natural logs and omitting to bring e into the work.

(ii) This part proved to be the most difficult question on the whole paper. Candidates failed to realise that the 'third second' is that from $t = 2$ to $t = 3$. Most substituted $t = 3$ into the equation for s and stopped. Of those who did grasp the idea some erred by using $t = 3$ and $t = 4$ (the fourth second) while others worked to just 3 figures and had an inaccurate answer.

(iii) Inappropriate use of 'speed = distance/time' gave $v = 1.6094... \div 2$, close enough to 0.8 to convince these candidates that they were on the right lines. However as the body did not have uniform speed this formula did not apply. All that was needed here was differentiation and substitution of $t = 2$.

(iv) Candidates who realised that acceleration equalled $\frac{dv}{dt}$ were now faced with the differentiation of a product or a quotient depending on how they chose to work. Relatively few quoted either formula and it was often not clear what candidates were doing.

Answers: **(i)** $t = 12.1$ **(ii)** ln2 m (or 0.693 m) **(iv)** $-\frac{6}{25}$ ms^{-2}

Question 11

Some candidates handled this question extremely well, but there were signs that others found this topic difficult. Inappropriate and incorrect substitutions were made, the most common being sin$y$ = –cos$y$ in part **(ii)**. Some candidates offered only one solution per part while others offered four. Candidates often lost accuracy marks for not following the instructions on the front of the question paper. In both part **(i)** and part **(ii)** answers were often correct to 3 significant figures rather than the 1 decimal place required for angles in degrees. In each case this had no effect on the first solution but did affect the second. In part **(iii)**, candidates who gave answers as decimals often did so to 2 decimal places rather than the generally required 3 significant figures.

(i) Most candidates knew to use tan$x$ although some made a sign error or had tan$x = \frac{3}{4}$. The second possible solution was often wrongly given as 233°.

(ii) The usual method here was to form a quadratic equation in sin$y$ using the identity $\cos^2 y = 1 - \sin^2 y$. Again the second solution was often given, inaccurately, as 166°.

(iii) This proved troublesome for many candidates. Dealing with the secant function was difficult for some, while others could not handle the bracket correctly. Of those who realised that $\cos\left(2z + \frac{\pi}{3}\right) = -\frac{1}{2}$ some could not decide whether to use 1.047 or 2.094 as their reference angle.

The operations of $-\frac{\pi}{3}$ then $\div 2$ were often carried out correctly but the first answer was frequently given as 0.52 and the second often omitted. Those candidates happy to work in multiples of π throughout produced, in general, short and correct solutions.

Some candidates worked the question in degrees throughout obtaining 30° and 90°, some then converting to radians, others forgetting. A few mixed degrees and radians in their working.

Answers: **(i)** x = 53.1°, 233.1° **(ii)** y = 14.5°, 165.5° **(iii)** $z = \frac{\pi}{6}, \frac{\pi}{2}$ (or 0.524, 1.57)

Question 12

12 EITHER

(i) As with **Question 11(iii)**, some candidates failed to deal with radians, treating the angles as being in degrees. Of those who made progress it was through differentation of the equation and then substituting as appropriate into the equations for y and for $\frac{\mathrm{d}y}{\mathrm{d}x}$. The value of A came out directly, and B by substitution into the original equation. Others were unable to find the value of A but used it correctly to find B.

(ii) This was well done, with candidates realising the need to integrate, which the majority did well, and to apply limits. Most evaluated at x = 0 rather than assuming the value of the integral at 0 is 0.

Answers: **(i)** $B = \sqrt{2}$ (or 1.41) **(ii)** Area = 3

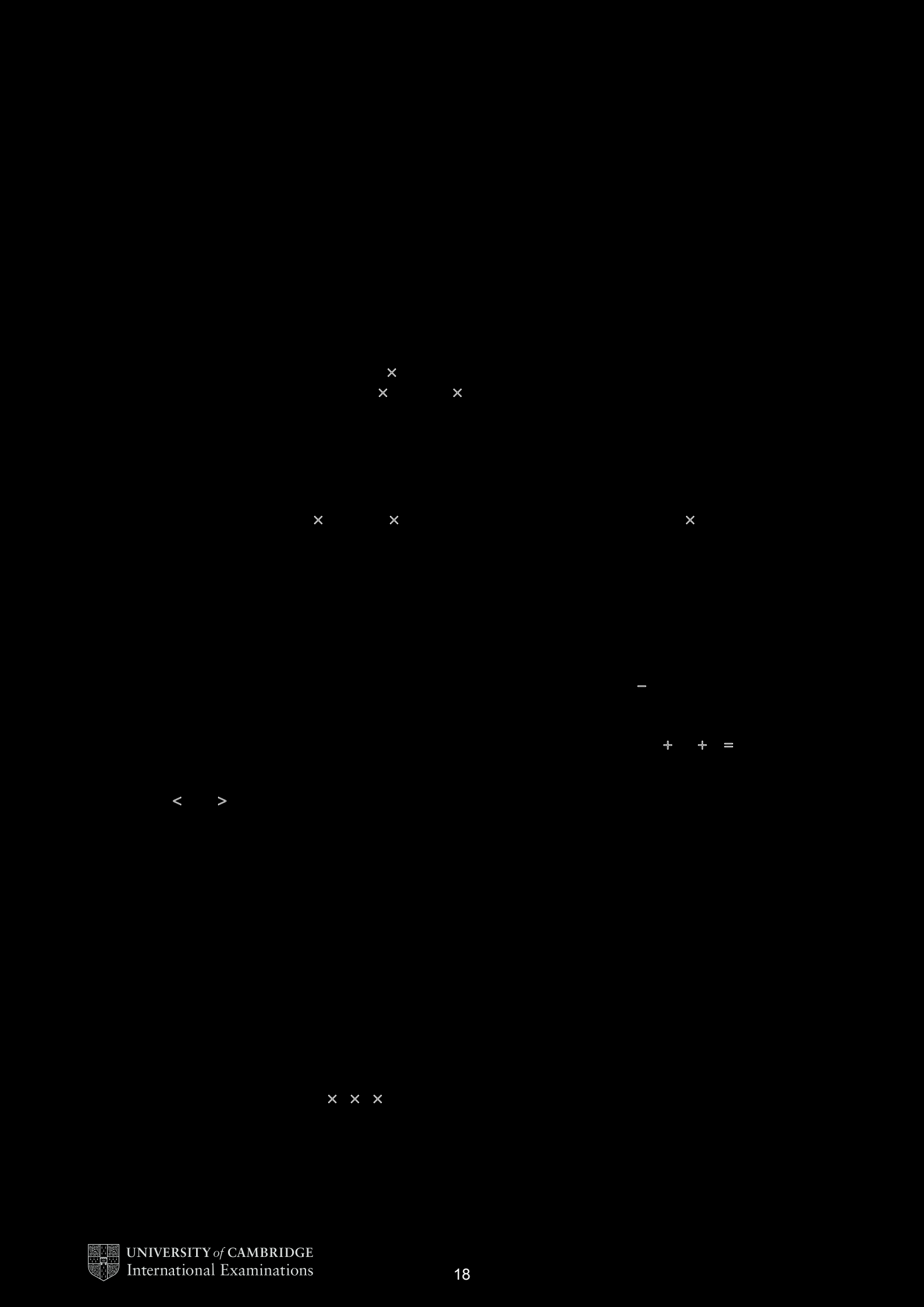
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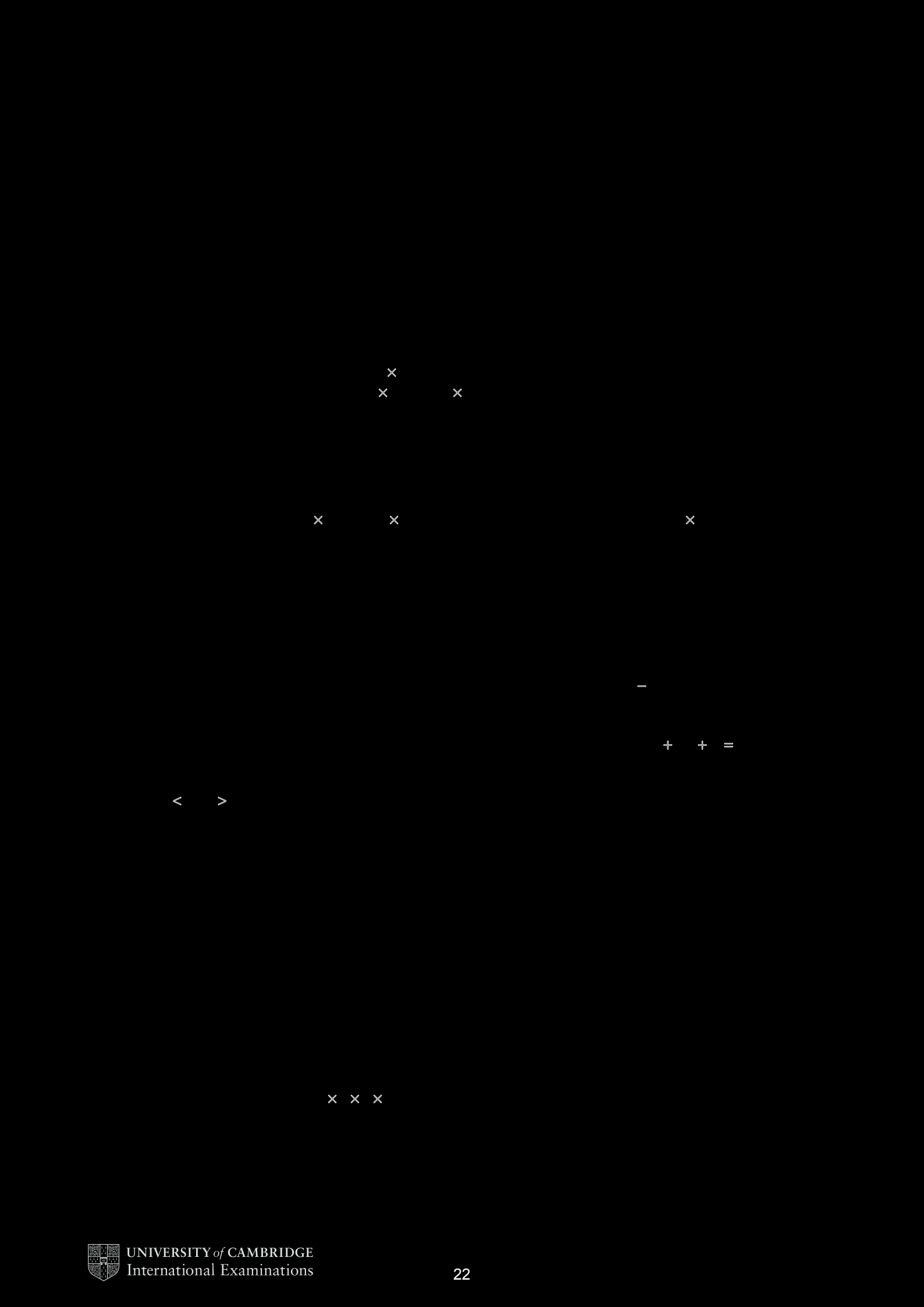
The majority of candidates answered this question.

(i) Attempts at this question covered the whole mark range. Some candidates knew that the normal had a gradient which was the negative reciprocal of something but only some of them realised the need to differentiate. Some did not manage to work out the coordinates of point *A* and thus had no point to put into their equation of the normal, or put point *C* into this which led nowhere. However a reasonable number did manage to obtain the required result.

(ii) The idea of integrating for the curve was used by almost all, and carried out correctly in most cases. The use of limits was more problematical. Some candidates used an upper limit of 2.5 (presumably the *y* value at point *C*) rather than 2 (the *x* value at *B*). Others failed to grasp the logic needed and integrated with a lower limit of 0. This left them the almost impossible task of finding area *OAC*. Some did recover from this, but many did not. It was expected that candidates would find the area under the curve between $x = 1$ and $x = 2$ and add this to the area of the trapezium with parallel sides *OC* and *AD* where *D* is directly below *A* on the *x*-axis. Only the better candidates seemed to cope with this.

Answers: **(ii)** $\frac{49}{12}$ (or 4.08).





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